

Semester 5

Guru Nanak Dev Engineering College, Ludhiana
An Autonomous College under UGC Act 1956
B.Tech. (CSE), Scheme - 2024

Course Code: CCS110
Course Title: Theory of Computation

Programme: B.Tech.	L: 3 T: 1 P: 0	Credits: 4
Semester: 5	Theory/Practical/ Integrated: Theory	Teaching Hours: 45(L)+15(T) = 60 hrs
Total Max. Marks: 100	Continuous Assessment (CA) Marks: 40	End Semester Examination (ESE) Marks: 60
Minimum Percentage of Numerical / Design / Programming Problems in ESE: 30%		
Duration of End Semester Examination (ESE): 3 hours		
Course Type: Core Course		

Prerequisites (if any): NIL

Additional Material Allowed in ESE: NIL

On completion of the course, the student will have the ability to:

CO#	Course Outcomes
1	Illustrate and analyze the usage of different types of finite automata (DFA, NFA, Moore and Mealy machines), and apply appropriate transformations to solve automata-based problems.
2	Explain the relationships among formal languages, grammar types, and language classes using the Chomsky hierarchy.
3	Apply the concepts of regular grammars, context-free grammars, and finite automata for language recognition and its simplification.
4	Design pushdown automata based on their computational capabilities to recognize and generate context-free languages.
5	Apply the principles of Turing machines to design computational models for solving complex engineering and decision problems.
6	Analyze and compare the computational capabilities of linear bounded automata with Turing machines in the context of context-sensitive languages.

Contents

Part-A

Unit-1 Finite Automata

7(L) hrs

Definitions of Finite State Machine, Transition systems, Deterministic Finite Automata (DFA), Acceptance by Finite Automata, Non-Deterministic Finite Automata, Equivalence of DFA and NDFA,

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Moore and Mealy machines, Equivalence of Moore and Mealy machine, Minimization of Finite Automata, Applications and Limitations of Finite Automata.

Unit-2 Formal Languages

5(L)hrs

Formal Languages and Regular Grammar: Basics of Alphabets, Strings, Language, Regular Expression, Grammar, Formal language, Chomsky classification of languages, Languages and their relation, Operations on languages, Closure properties of language classes.

Unit-3 Regular Grammar

8(L) hrs

Regular grammars, Regular expressions, Operators of Regular Expression and their precedence, Algebraic Laws for Regular Expressions, Kleen's Theorem, Regular Expression to Finite Automata, DFA to Regular Expression -Algebraic method using Arden's Theorem, Equivalence of Finite Automata and Regular expressions, Properties of Regular Languages, Pumping Lemma

Part-B

Unit-4 Context Free Language

8(L)hrs

Derivation Tree, Ambiguity, Simplification of Context Free Grammar, Normal forms– Chomsky Normal Form, Greibach Normal Form, Pumping lemma.

Unit-5 Push Down Automata

8(L)hrs

Description and Definition, Acceptance by Push Down Automata, Equivalence of Push Down Automata and Context Free Grammars and Languages.

Unit-6 Turing Machine Linear Bounded Automata

9(L) hrs

Definition and Model, Representation of Turing Machine, Design of Turing Machine, Variants of Turing Machine, Decidability and recursively enumerable languages, Halting problem, Post correspondence problem, Context sensitive language, Model of linear bounded automata, Relation between linear bounded automata and context sensitive language.

Text Books

1. Hopcroft, J. E., Motwani, R., & Ullman, J. D. "Introduction to Automata Theory, Languages, and Computation", 3rd edition, Pearson Education India, 2008.
2. Mishra, K. L. P., & Chandrasekaran, N. "Theory of Computer Science: Automata, Languages and Computation", 3rd edition, PHI Learning Pvt. Ltd. 2011.

Reference Books

1. Sunitha, K. V. N., & Kalyani, 'N. Formal Languages and Automata Theory", 1st edition, Pearson, 2015.
2. Daniel, A.Cohen, "Introduction to Computer Theory", 2nd edition, John Wiley& sons Inc.

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3. M. Sipser, “Introduction to the Theory of Computation”, 3rd edition, Cengage Learning, 2013.
4. Peter Linz, “An Introduction to Formal Languages and Automata”, 6th edition, Jones and Bartlett Publishers, 2017.

E-Books and Online Learning Material

1. Jean-Eric Pin, Mathematical Foundations of Automata Theory, Lecture notes LIAFA, Université Paris <https://www.irif.fr/~jep//PDF/MPRI/MPRI.pdf>
2. Michael Sipser, Introduction to the Theory of Computation, Thomson Course Technology <http://en.bookfi.net/book/1139836>
3. Anil Maheshwari and Michiel Smid, Introduction to Theory of Computation, School of Computer Science, Carleton University, Ottawa Canada <https://cglab.ca/~michiell/TheoryOfComputation/TheoryOfComputation.pdf>
4. Michael Levet, Theory of Computation, Lecture Notes, University of South Carolina-Columbia http://people.math.sc.edu/mlevet/Lecture_Notes.pdf

Supplementary SWAYAM Course

Sr. No.	Course Name	Instructor	Host Institute	URL
1.	Theory of Computation	Prof. Raghunath Tewari	IIT Kanpur	https://nptel.ac.in/courses/106104148
2.	Theory of Computation	Prof. Somenath Biswas	IIT Kanpur	https://nptel.ac.in/courses/106104028
3.	Theory of Automata, Formal Languages and Computation	Prof. Kamala Krithivasan	IIT Madras	https://nptel.ac.in/courses/106106049

Guru Nanak Dev Engineering College, Ludhiana
An Autonomous College under UGC Act 1956
B.Tech. (CSE), Scheme - 2024

Course Code: CCS111
Course Title: Cyber Security

Programme: B.Tech.	L: 3 T: 0 P: 2	Credits: 4
Semester: 5	Theory/Practical/ Integrated: Integrated	Teaching Hours: 45(L) + 30 (P) = 75 hrs
Total Max. Marks: 150	Continuous Assessment (CA) Marks: 90	End Semester Examination (ESE) Marks: 60
Minimum Percentage of Numerical / Design / Programming Problems in ESE: 20%		
Duration of End Semester Examination (ESE): 3 hours		
Course Type: Core Course		

Prerequisites (if any): CCS106

On completion of the course, the student will have the ability to:

CO#	Course Outcomes
1	Apply cyber security principles within legal, regulatory, and ethical frameworks of cyberspace.
2	Analyze cyber security threats and attacks impacting systems, networks, and applications.
3	Explore cyber security strategies, risk management practices, and emerging approaches to strengthen organizational security.
4	Examine security protocols and defence mechanisms for secure communication and protection.
5	Evaluate ethical, social, and human factors influencing cyber security practices and incidents.
6	Design preventive and response measures based on real-world cyber security incidents.

Contents

Part A

Unit 1 Introduction to Cyber Security

7 (L)hrs

Definition and scope of cyber security and cybercrime, classification of cybercrimes, relationship between cybercrime and cyber security, legal and regulatory frameworks for cyber security, IT Act (2000), IT Act Amendments (2008) and key sections, global perspective on cybercrime and cyber security, cyber security awareness, and strategies for safe and responsible use of cyberspace.

Unit 2 Cyber Security Threats and Attacks

7 (L)hrs

Overview of Cyber Security Threats and Attacks, Proxy servers and anonymizers, phishing attacks, password cracking techniques, keyloggers and spyware, viruses and worms, Trojan horses and

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backdoors, steganography, denial of service and distributed denial of service attacks, SQL injection attacks, buffer overflow attacks, ransomware and cryptojacking, exploit kits and zero-day attacks.

Unit 3 Cyber Security Strategies and Emerging Technologies

7 (L)hrs

Security governance and frameworks (ISO 27001, NIST), risk management and assessment, security policies and standards, business continuity and disaster recovery planning, security auditing and compliance, cloud computing security challenges, mobile and wireless device security, BYOD, Fundamentals of Defence-in-Depth Security, Basics of Zero Trust Security Model, security-by-design, DevSecOps Principles, Quantum security and Post-Quantum security Awareness.

Part B

Unit 4 Protocols and Techniques in Cyber Security

11 (L)hrs

Fundamentals of security protocols and secure communication, conceptual study of SSL/TLS for protecting data in transit, functional aspects of IPsec in network-layer security, working principles of SSH for secure remote system access, role of HTTPS in securing web-based communication, fundamental mechanism of Kerberos authentication, essential features of secure email systems (PGP and S/MIME), core ideas behind wireless security standards (WPA, WPA2, WPA3), common challenges and limitations of widely used security protocols, network security techniques (firewalls, IDS/IPS), malware detection and prevention techniques, vulnerability assessment and penetration testing, incident response and recovery techniques, security monitoring, logging, and policy implementation.

Unit 5 Cyber Ethics and Human Factors in Cyber Security

7 (L)hrs

Cyber ethics principles, ethical issues in cyberspace, privacy and data protection concerns, psychological aspects of attackers, user and victim behavior, principles and responsibilities of ethical hacking, social engineering awareness and prevention, organizational security culture and human responsibility.

Unit 6 Case Studies

6 (L)hrs

Real-world cyber security case studies, analysis of cyber-attacks, lessons learned from incidents, preventive strategies, legal and ethical implications of major cyber security incidents.

Laboratory Work

In the laboratory, students will perform hands-on practical exercises using Wireshark, web browser-based tools etc. to analyze cyber security concepts. Laboratory experiments are aligned with theory and designed for real-world applications.

The following is only the suggested list of Practical's. Instructors may frame additional Practical's relevant to the course contents.

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Experiment No.	Experiment Title
1.	Study of Cybercrime Scenarios and Associated Impacts.
2.	Study a cybercrime case related to the IT Act, 2000 and identify applicable sections.
3.	Analysis of Active Phishing Attacks and Identification of Fake Websites.
4.	Password Strength Evaluation and Analysis Using Browser-Based Tools.
5.	Installation and Configuration of Wireshark on Windows and Linux Platforms
6.	Observation and Analysis of Active Denial-of-Service Attack Traffic Patterns Using Wireshark.
7.	Comparative Analysis of Secure and Insecure Web Communication (HTTP vs HTTPS).
8.	Packet Capture and Network Traffic Analysis Using Wireshark.
9.	Observation and Analysis of SSL/TLS Encrypted Network Traffic.
10.	Protocol Analysis of HTTP, DNS, and TCP Using Wireshark.
11.	Network Log Monitoring and Incident Detection Using Wireshark.
12.	Study of Social Engineering Techniques and Cyber Ethics Awareness.
13.	Case Study Analysis of Real-World Cyber Security Incidents

Text Books

1. Sunit Belapure and Nina Godbole, “Cyber Security: Understanding Cyber Crimes”, Computer Forensics and Legal Perspectives, Wiley India, 1st edition, 2011.
2. Thomas A. Johnson, “Cyber Security”, CRC Press, 1st edition, 2015.
3. Eric Cole, “Cyber Security Essentials”, Wiley, 1st edition, 2010.

Reference Books

1. William Stallings and Lawrie Brown, “Computer Security: Principles and Practice”, Pearson Education, 4th edition, 2018.
2. Charles P. Pfleeger and Shari Lawrence Pfleeger, “Security in Computing”, Prentice Hall, 5th edition, 2015.

Online Learning Material

1. <https://www.coursera.org/learn/introduction-to-cybersecurity-fundamentals>

Accessed on 14-12-2025

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| 2. https://www.nist.gov/cyberframework | Accessed on 14-12-2025 |
| 3. https://owasp.org/www-project-top-ten/ | Accessed on 14-12-2025 |
| 4. https://www.cloudflare.com/learning/security/ | Accessed on 14-12-2025 |
| 5. https://www.cisa.gov/cybersecurity | Accessed on 14-12-2025 |

Supplementary SWAYAM Course

Sr. No.	Course Name	Instructor	Host Institute	URL
1	Introduction to Cyber Security	Dr. Jeetendra Pande	Uttarakhand Open University (via SWAYAM)	https://onlinecourses.swayam2.ac.in/nou19_cs08/preview
2	Cyber Security	Dr. G. Padmavathi	Avinashilingam Institute (via SWAYAM)	https://onlinecourses.swayam2.ac.in/cec23_cs16/preview
3	Cyber Security, Tools, Techniques and Counter Measures	Prof. Nilesh K. Modi	Dr. Babasaheb Ambedkar Open University (via SWAYAM)	https://onlinecourses.swayam2.ac.in/nou25_ge62/preview

Guru Nanak Dev Engineering College, Ludhiana
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B.Tech. (CSE), Scheme - 2024

Course Code: CCS112
Course Title: Software Engineering

Programme: B.Tech.	L: 3 T: 0 P: 2	Credits: 4
Semester: 5	Theory/Practical/ Integrated: Integrated	Teaching Hours: 45(L) + 30 (P) = 75 hrs
Total Max. Marks: 150	Continuous Assessment (CA) Marks: 90	End Semester Examination (ESE) Marks: 60
Minimum Percentage of Numerical / Design / Programming Problems in ESE: 5%		
Duration of End Semester Examination (ESE): 3 hours		
Course Type: Core Course		

Prerequisites (if any): NIL

On completion of the course, the student will have the ability to:

CO#	Course Outcomes
1	Apply various software process models and agile methods to software development problems.
2	Analyze requirements and represent decision logic using techniques like decision trees and tables.
3	Create design diagrams such as DFDs, ER diagrams and structure charts based on software design principles.
4	Evaluate project management tools including metrics, estimation methods, scheduling and risk plans.
5	Apply testing techniques including unit, integration and system testing to various software problems.
6	Analyze version control systems and software maintenance approaches for managing software evolution.

Contents

Part-A

Unit-1 Introduction to Software Engineering

3(L) hrs

Nature of Software, Software Engineering, Software Process, Software Myths, Software application domains, Software crisis – Problem and causes.

Unit-2 Software Development Lifecycle

10(L) hrs

Software Process Models - Waterfall model, Prototype model, Spiral model, Incremental model, Evolutionary model, Agile model. Introduction to Software Development Methodologies – Agile development - Scrum, XP. DevOps principles (CI/CD).

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Unit-3 Requirement Engineering

9(L) hrs

Problem Analysis, Requirement Engineering Process, Requirement Elicitation, Requirement Specification and Validation, documenting Software Requirement Specification (SRS), Requirement Analysis Approaches- Structured Analysis vs Object Oriented Analysis, Functional and non-functional requirements, Techniques for representing complex logic – Decision trees, Decision tables.

Part-B

Unit-4 Software Project Management

8(L) hrs

Project Management Concepts – Managing People, Product, Process and Project. Process and Project Metrics – Size oriented metrics, Function oriented metrics, Empirical estimation, Heuristic estimation and Analytical estimation, Project monitoring and control– Work breakdown structure, Activity chart, Gantt charts, PERT charts, Critical Path Method, Risk management- Identification, Analysis, Planning and Monitoring, RMMM Plan.

Unit-5 Software Design and Coding

7(L) hrs

Software Design Principles, Software design concepts – Abstraction, Architecture, Modularity, Functional Independence, Refinement, Refactoring, Function oriented design – Flow modeling through Data Flow Diagram and Data Dictionary, Data Modeling through E-R Diagram, Structure chart, Object-Oriented Design Concepts – objects and object classes. Coding standards, code inspection, code walkthrough and code reviews.

Unit-6 Software Testing and Maintenance

8(L) hrs

Introduction to Software Testing, Types of Software Testing - Unit Testing, Integration Testing, System Testing, Test Planning and Test Cases, Test Automation, Software Verification and Validation, Software maintenance, Software re-engineering, Reverse engineering, Forward engineering, Challenges in Legacy Systems, Software Updates and Version Control.

Laboratory Work

In the laboratory work, students should implement these concepts using any open-source tools. Students are also required to choose a suitable project and perform the following practical's on the same.

Following is only the suggested list of Practical's. Instructor may frame additional Practical's relevant to the course contents.

Experiment No.	Experiment Title
1.	Description of the project (Title, Introduction, Purpose etc.)
2.	Conduct feasibility study for the project.
3.	Create a Software Process Lifecycle Diagram using tools like draw.io/Lucid chart etc. for the project chosen and justify why the chosen SDLC model is appropriate.

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4.	Preparation of Software Requirement Specification (SRS) document by clearly specifying all functional and non-functional requirements.
5.	Draw Decision Trees and Tables using draw.io or similar software for a sample logic problem of the project.
6.	Study of OpenProj / Project Libre or similar software for Software Project Management.
7.	Implement Activity diagram and Resource usage of the project using OpenProj / Project Libre or similar software.
8.	Implement Gantt Chart of the project using OpenProj / Project Libre or similar software. Also find critical path.
9.	Study and implement any function-oriented design tool(s) for making Data Flow Diagrams (DFD).
10.	Create ER diagram for the project problem using any open-source tool like draw.io/ dbdiagram.io etc.
11.	Study and implement any function-oriented design tool(s) for making Structure Chart.
12.	Compute test cases using equivalence class partitioning and boundary value analysis for the project sub-problems.
13.	Study of testing tools like Selenium/ Appium/ Postman etc. to understand the testing process.
14.	Perform functional testing of any website like IRCTC, bookmyshow etc. to test search/login and other features and document the observations.

Text Books

1. Roger Pressman and Bruce Maxim, “Software Engineering: A Practitioner’s Approach”, 9th edition by, McGraw-Hill Education, 2023.
2. Rajib Mall, “Fundamentals of Software Engineering”, Prentice Hall of India, 5th edition, 2018.
3. Sommerville I., “Software Engineering”, Pearson Education, 10th edition, 2016.

Reference Books

1. P. Jalote, “An Integrated Approach to Software Engineering”, Springer, 3rd edition, 2005.
2. B. Hughes, M. Cortell, R. Mall, “Software Project Management”, Tata McGraw Hill, 5th edition, 2009.

Online Learning Materials

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| 1. https://nptel.ac.in/courses/10610508 | Accessed on 20-05-2025 |
| 2. https://www.scrum.org/learning-series/what-is-scrum/ | Accessed on 20-05-2025 |
| 3. https://www.guru99.com/software-testing.html | Accessed on 20-05-2025 |
| 4. https://www.youtube.com/watch?v=T0TynxN77oY | Accessed on 20-05-2025 |
| 5. https://www.youtube.com/watch?v=LA_N8cvmLBI | Accessed on 20-05-2025 |

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Supplementary SWAYAM Course

Sr. No.	Course Name	Instructor	Host Institute	URL
1	Software Engineering	Prof. Rajib Mall	IIT Kharagpur	https://onlinecourses.nptel.ac.in/no_c20_cs68/preview
2	Software Project Management	Prof. Rajib Mall, Prof. Durga Prasad Mohapatra	IIT Kharagpur, NIT Rourkela	https://onlinecourses.nptel.ac.in/no_c22_cs107/preview
3	Software Engineering	Prof. N.L. Sarda, Prof. Umesh Bellur, Prof. R.K. Joshi	IIT Bombay	https://nptel.ac.in/courses/106101061

Guru Nanak Dev Engineering College, Ludhiana
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B.Tech. (CSE), Scheme - 2024

Course Code: CCS113
Course Title: Design and Analysis of Algorithms

Programme: B.Tech.	L: 3 T: 0 P: 2	Credits: 4
Semester: 5	Theory/Practical/ Integrated: Integrated	Teaching Hours: 45(L)+30(P) = 75 hrs
Total Max. Marks: 150	Continuous Assessment (CA) Marks: 90	End Semester Examination (ESE) Marks: 60
Minimum Percentage of Numerical / Design / Programming Problems in ESE: 90%		
Duration of End Semester Examination (ESE): 3 hours		
Course Type: Core Course		

Prerequisites (if any): CCS102

On completion of the course, the student will have the ability to:

CO#	Course Outcomes
1	Evaluate the time and space complexity of recursive and non-recursive algorithms.
2	Apply divide-and-conquer strategies to construct efficient algorithmic solutions.
3	Implement Breadth-First Search (BFS) and Depth-First Search (DFS) techniques for solving graph-based problems.
4	Solve optimization problems using appropriate greedy algorithm strategies.
5	Design Dynamic Programming solutions to solve complex optimization problems.
6	Optimize solution strategies for NP-complete problems using reductions and approximation algorithms.

Contents

Part-A

Unit-1 Introduction

4(L) hrs

Algorithm design and representation, Performance analysis- Space complexity, Time complexity, Asymptotic Notations- Big-Oh (O), Omega (Ω), Theta (Θ), and Little-oh (o), Little Omega (ω) Notation, Mathematical analysis of non-recursive and recursive algorithms with examples.

Unit-2 Divide and Conquer

5(L) hrs

General method, solving recurrences using recurrence trees, repeated substitution, statement of Master Theorem, applications – Binary search, Merge sort, Quick sort, Strassen's Matrix Multiplication, Finding the maximum and minimum.

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Unit-3 Applications of Graph Traversal Techniques

6(L)hrs

Representation of graphs, Breadth-First Search method for Single Source Shortest Path on unweighted graphs, Depth First Search using queue, types of connected, biconnected and strongly connected components in directed graphs, topological sorting using Directed Acyclic Graph (DAG).

Unit-4 Greedy Algorithms

7(L)hrs

Greedy choice, optimal substructure property, minimum spanning trees-Prim's and Kruskal's, shortest path using Dijkstra, Bellman Ford, fractional knapsack, Job Sequencing with Deadline, Activity Selection Problem, Travelling salesperson problem and Huffman coding.

Part-B

Unit-5 Dynamic Programming

7(L)hrs

Introduction to dynamic programming and application of the algorithm to solve multistage graphs with Forward and backward approach, edit distance, matrix chain multiplication, All pairs shortest path problem using Floyd-Warshall algorithm and 0/1 Knapsack problem.

Unit-6 String Matching

5(L)hrs

Introduction, Brute Force algorithm, Rabin-Karp algorithm using Hashing Technique, KMP algorithm with π Table, Boyer-Moore algorithm using Bad Match table and Good Suffix heuristic.

Unit-7 Backtracking

5(L)hrs

General method, N-Queens problem, Sum of subsets problem, Graph coloring, Hamiltonian cycles.

Unit-8 NP Completeness

6(L)hrs

Classes NP, P, NP- Hard, NP-complete and polynomial time reductions, Introduction to approximation algorithms, Absolute approximations, E-approximations, Travelling salesperson problem with NP-complete, SAT (Boolean Satisfiability Problem) to IND-SET (Independent Set Problem) reduction. IND-SET to Vertex Cover, 2-approximation algorithm for Vertex Cover.

Laboratory Work

The following is only the suggested list of Practical's. Instructors may frame additional Practical's relevant to the course contents.

Experiment No.	Experiment Title
1	A college library system stores roll numbers in both sorted and unsorted formats for different departments. Analyze how the choice between Linear Search and Binary Search affects search efficiency in each case.

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2	Write a program to sort the class roll numbers of your class using a merge sort algorithm and determine the time required to sort the elements.
3	Write a program to sort the university roll numbers of your class using the Quick sort method and determine the time required to sort the elements.
4	Write a program to use a queue to store the node and mark it as 'visited' until all its neighbours (vertices that are directly connected to it) are marked. Implement by using BFS and DFS algorithms for a graph.
5	Write a program to solve fractional knapsack using the Greedy algorithm.
6	Implement Prim's and Kruskal's algorithms for a common network and compare their effectiveness in finding the minimum spanning tree.
7	Design a program that models a road network and finds the shortest path from home to college using Dijkstra's and Bellman-Ford algorithms.
8	Write a program to find the shortest path of the multistage graph using dynamic programming.
9	Implement the Floyd-Warshall algorithm to compute shortest paths among cities in a state-wide network.
10	Write a program to match the pattern by using Brute Force algorithm and Rabin-Karp algorithm.
11	Write a program to match the String by using KMP algorithm and Boyer-Moore algorithm.
12	Write a program to find the solution to the 8-queens problem using backtracking.
13	Write a program to solve subset sum problems using backtracking.

Text Books

1. Ellis Horowitz, Sartaj Sahni and S. Rajasekharan, "Fundamentals of Computer Algorithms", Universities Press, 2nd Edition, 2007.
2. T. H. Cormen, C. E. Leiserson, R. L. Rivest, and C. Stein, "Introduction to Algorithms", PHI, 3rd Edition, 2009.
3. P. H. Dave, H. B. Dave, "Design and Analysis of Algorithms", Pearson Education, 2nd Edition, 2007.

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Reference Books

1. M. T. Goodrich and R. Tomassia, “Algorithm Design: Foundations, Analysis and Internet examples”, John Wiley and sons, 1st Edition, 2001.
2. S. Sridhar, “Design and Analysis of Algorithms”, Oxford Univ. Press, 2nd Edition, 2014.
3. Aho, Ullman and Hopcroft, “Design and Analysis of algorithms”, Pearson Education, 2nd Edition, 2004.
4. R. Neapolitan and K. Naimipour, “Foundations of Algorithms”, Jones and Bartlett Student edition, 4th edition, 2010.

Online Learning Materials

1. <https://www.coursera.org/learn/cpsc-8400-design-and-analysis-of-algorithms>
Accessed on Dec 27, 2025
2. <https://www.classcentral.com/course/youtube-computer-sc-design-analysis-of-algorithms-47664>
Accessed on Dec 27, 2025

Supplementary SWAYAM Course

Sr. No.	Course Name	Instructor	Host Institute	URL
1	Design and analysis of algorithms	Prof. Madhavan Mukund	Chennai Mathematical Institute	https://onlinecourses.nptel.ac.in/noc26_cs67/preview
2	Fundamental Algorithms: Design and Analysis	Prof. Sourav Mukhopadhyay	IIT Kharagpur	https://onlinecourses.nptel.ac.in/noc26_cs42/preview
3	Design and Analysis of algorithms	Dr. Faheem Syeed Masoodi	University of Kashmir	https://onlinecourses.swayam2.ac.in/cec20_cs03/preview

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B.Tech. (CSE), Scheme - 2024

Course Code: ESCS102
Course Title: Design Thinking

Programme: B.Tech.	L: 0 T: 0 P: 2	Credits: 1
Semester: 5	Theory/Practical/Integrated: Practical	Teaching Hours: 30(P)=30hrs
Total Max. Marks: 50	Continuous Assessment (CA) Marks: 50	End Semester Examination (ESE) Marks: 0
Minimum Percentage of Numerical / Design / Programming Problems in ESE: 0		
Duration of End Semester Examination (ESE): NIL		
Course Type: Core Course		

Prerequisites (if any): NIL

On completion of the course, the student will have the ability to:

CO#	Course Outcomes
1	Identify user-centric problems using Design Thinking principles.
2	Apply Design Thinking tools to empathize with users and define problems.
3	Generate and evaluate innovative ideas using structured ideation techniques.
4	Develop and refine prototypes through iterative Design Thinking models.
5	Present ethical and sustainable solutions using effective storytelling and documentation.

Contents

Unit-I Overview of Design Thinking

4(P) hrs

Introduction to Design Thinking- drivers, attributes, principles of design thinking, 3Ps of design thinking, features and applications of design thinking, design thinking vs traditional thinking. Case Studies- General, Engineering and Service applications.

Activities: Identify an Opportunity and Scope of the Project Explore the possibilities and Prepare design brief

Unit-II Design Thinking tools and techniques

7(P) hrs

Design Thinking persona, visualisation, stakeholder mapping, journey mapping, mind mapping, Methods and Tools for Empathize and Define phases, star bursting, divergent thinking, convergent thinking, ethnography, brainstorming, storytelling, role playing and user interviews. Six thinking hats.

Activities: Apply Design Thinking Tools and Techniques to Solve a User-Centric Problem.

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Unit-III Design Thinking Models

12 (P) hrs

Double diamond design thinking model- 4D's of double diamond (Discover, Define, Develop and Deliver), feedback mechanism. Stanford 5-Phase design thinking model- Empathize, Define, Ideate, Prototype and Test. Feedback and iterations.

Activities:

- Apply the methods of empathize and Define Phases Finalize the problem statement
- Apply the methods of Ideate Phase: Generate lots of Ideas
- Apply the methods of Prototype Phase: Create prototypes for selected ideas
- Collect feedback; iterate and improve the ideas
- Present your solution using Storytelling method

Unit-IV Ethics in Design Thinking and Design Thinking for Sustainability

3 (P)hrs

Design Thinking Ethics- values and morals, five ways of ethical thinking, ethical issues, universal principles of ethics. Sustainability- three pillars of sustainability, sustainable design, 2030 agenda for sustainable development.

Project Submission: Fine tuning and submission of project report

4 (P)hrs

Note: Students shall form into groups and identify a problem (preferably societal problem with engineering orientation to solve) suitable for the design thinking and go through the process week-wise. At the end of each phase, brief documentation shall be submitted and a final report covering all phases has to be submitted at the end of the semester.

Text Books:

1. E Balagurusamy, B. Vijaya Kumar, "Design Thinking A Beginners Perspective", McGraw Hill Education India, 1st Edition, 2025.
2. Michael Lewrick, Patrick Link, Larry Leifer, "The Design Thinking Playbook: Mindful Digital Transformation of Teams, Products, Services, Businesses and Ecosystems", Wiley, Edition- May 2018

Reference Books:

1. Michael Lewrick, Patrick Link, Larry Leifer, "The Design Thinking Toolbox: A Guide to Mastering the Most Popular and Valuable Innovation Methods", Wiley, Edition- March 2020
2. Shalini Rahul Tiwari, Rohit Rajendra Swarup, "Design Thinking: A Comprehensive Textbook", Wiley India, 2024.
3. Tim Brown , "Change by Design, Revised and Updated How Design Thinking Transforms Organizations and Inspires Innovation", Harper Collins, 2019

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Online Learning Material

1. Design Thinking and Innovation — IIT Bombay (Coursera)
<https://www.coursera.org/learn/designthinkingandinnovation> Accessed on 09-01-2026
2. Design Thinking for Innovation — University of Virginia (Coursera)
<https://www.coursera.org/learn/uva-darden-design-thinking-innovation> Accessed on 09-01-2026
3. Design Thinking and Innovation — Harvard Business School Online
<https://online.hbs.edu/courses/design-thinking-innovation/> Accessed on 09-01-2026

Supplementary SWAYAM Course

Sr. No.	Course Name	Instructor	Host Institute	URL
1	Design Thinking and Innovation	Prof. Ravi Poovaiah	IIT Bombay	https://onlinecourses.swayam2.ac.in/aic26_ge05/preview